COMPUTER NETWORKS (24MC410)

I M.C.A II Semester
2025-26 EVEN SEMESTER

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VFSTR (Deemed to be University)

Institute Vision & Mission

VIGNAN'S FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

VISION:

To evolve in to a centre of excellence in Science & Technology through creative and innovative practices in teaching-learning, towards promoting academic achievement and research excellence to produce internationally accepted, competitive and world class professionals who are psychologically strong & emotionally balanced imbued with social consciousness & ethical values.

MISSION:

To Provide High Quality academic programmes, training activities, research facilities and opportunities supported by continuous industry - institute interaction aimed at promoting employability, entrepreneurship, leadership and research aptitude among students and contribute to the economic and technological development of the region, state and nation.

Department Vision & Mission

VISION:

To emerge as a renowned center in the field of computer applications for providing high quality education and research to produce globally competent professionals with ethical values.

MISSION:

- **M1:** To provide high quality education with cutting-edge curriculum and learner-centered approach.
- **M2:** To establish the culture of innovation and research through industry- academia linkages.
- **M3:** To cultivate ethical and socially-conscious professionals with leadership qualities, who understand the global implications of technology.

PO's of the Department

- **PO1 (Foundation Knowledge):** Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.
- **PO2 (Problem Analysis):** Identify, review, formulate and analyze problems for primarily focusing on customer requirements using critical thinking frameworks.
- **PO3 (Development of Solutions):** Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.
- **PO4 (Modern Tool Usage**): Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

PO's of the Department

- **PO5 (Individual and Teamwork):** Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.
- **PO6 (Project Management and Finance)**: Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.
- **PO7 (Ethics):** Commit to professional ethics in managing software projects with financial | aspects. Learn to use new technologies for cyber security and insulate customers from malware
- **PO8 (Life-long learning):** Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

PEO's of the Department

- **PEO1:** Develop cutting-edge technical skills to succeed in career and as an entrepreneur.
- **PEO2 :** Pursue higher education and research with critical thinking and problem solving capabilities
- **PEO3:** Demonstrate ethical, collaborative, and leadership abilities in multi-disciplinary domains.

PSO's of the Department

- PSO1: To provide solutions for complex computational problems with requisite mathematical, programming languages and tools.
- PSO2: To design, develop and analyse software and related services using algorithms, web design, and Big Data Analytics for real time applications.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on imparting knowledge about various protocols involved in LANs and WANs. In addition, it gives good foundation on different protocols such as data link protocols, internet protocols and transport protocols present in the respective layers of computer networks.

Syllabus
Module-1

UNIT – II

PHYSICAL LAYER AND TRANSMISSION MEDIA

Guided Transmission Media: Twisted Pair, Coaxial, Fiber Optics, Wireless Transmission, Using the Spectrum for Transmission: Radio, Microwave, Infrared, Light transmission.

Guided Transmission Media

- The **physical layer** is responsible for transferring raw bits from one machine to another using a transmission medium.
- When this transmission uses a **physical path such as a cable or wire**, it is called **guided transmission media.**
- The most commonly used guided media are **copper cables** (including twisted pair and coaxial cables) and **optical fiber**.
- Each type of medium has different characteristics and trade-offs related to **bandwidth, frequency, delay, cost, and ease of installation and maintenance**.
- **Bandwidth**, which represents the data-carrying capacity of a medium, is measured in **Hertz (Hz)** and indicates how much information can be transmitted over the medium in a given time.

Persistent Storage

- One of the simplest and most common ways to transfer data between computers is by **copying the data onto persistent storage** such as magnetic tapes, hard disks, or solid-state media, physically transporting that storage to the destination, and then reading the data back.
- Although this method may seem old-fashioned compared to satellites or high-speed networks, it is often **far more cost-effective** when the goal is to move **very large volumes of data**.
- This approach is also extremely **cheap**. No network-based data transfer can match this cost efficiency.
- For extremely large data transfers, companies still use this method.

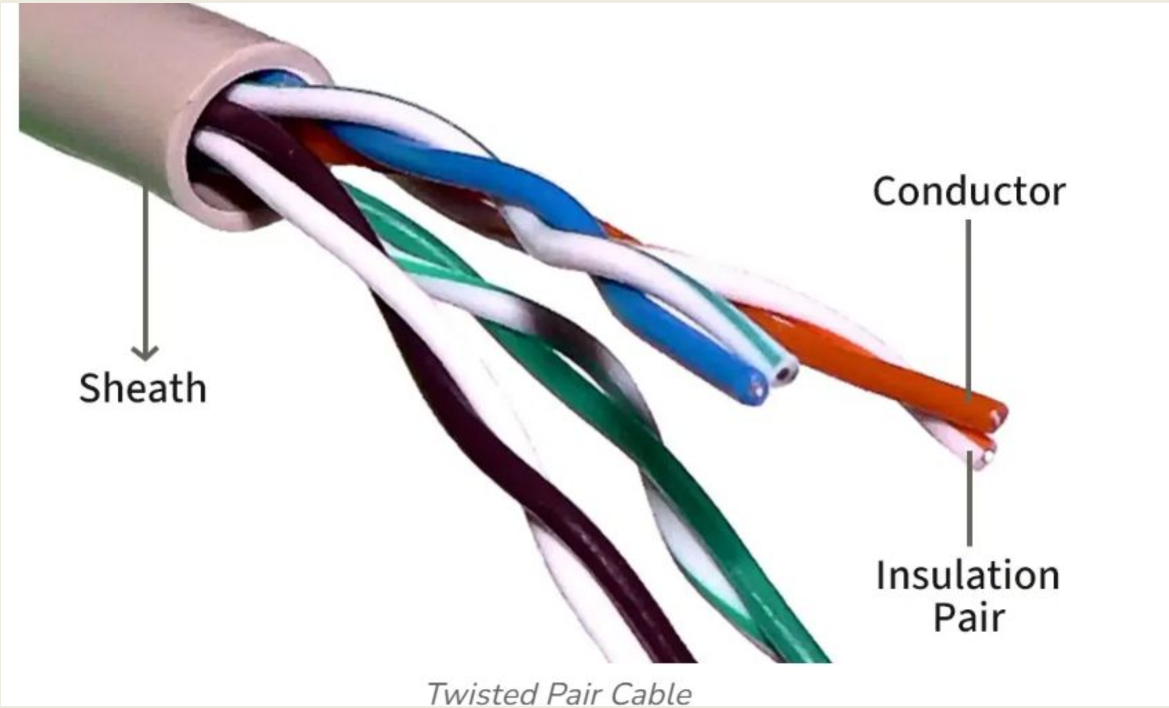
- For example, **Amazon's Snowmobile** is a massive truck filled with thousands of hard disks, capable of transporting up to **100 petabytes (PB)** of data.

- The truck connects directly to a company's network, copies the data, drives to another location, and unloads it.
- For organizations moving huge datasets, such as migrating an entire data center to the cloud, **physical transport remains the fastest and most practical solution**.

TWISTED-PAIR

- Although **persistent storage** (like disks) can carry large amounts of data, it has **very high delay**, with transmission times measured in **hours or days**.
- It unsuitable for real-time applications such as the **Web, video conferencing, and online gaming**, where **low delay** is essential.
- One of the oldest & most widely used guided transmission media is the **twisted-pair cable**.
- A twisted pair consists of **two insulated copper wires**, usually about **1 mm thick**, twisted together in a helical shape.
- Twisting the wires reduces interference because electromagnetic waves produced by one twist cancel those of the next, minimizing signal radiation and noise.

- In twisted-pair transmission, data is carried as the **difference in voltage between the two wires**, rather than an absolute voltage.
- This **differential signaling** improves resistance to external noise, since interference typically affects both wires equally, leaving the voltage difference largely unchanged.



- The most common use of twisted-pair cables is in the **telephone system**, where almost all telephones connect to the telephone company using twisted pairs

- These same lines also support **ADSL Internet access**. Twisted pairs can transmit signals over **several kilometers without amplification**, but over longer distances, the signal weakens and **repeaters** are required.

Two Conductors Twisted Together

- When many twisted pairs run together, such as in cables connecting apartment buildings to telephone exchanges, they are **bundled and encased in protective sheaths**.

- Without twisting, these closely packed wires would interfere with each other.
- Due to their **low cost, widespread availability, and acceptable performance**, twisted pairs continue to be widely used.

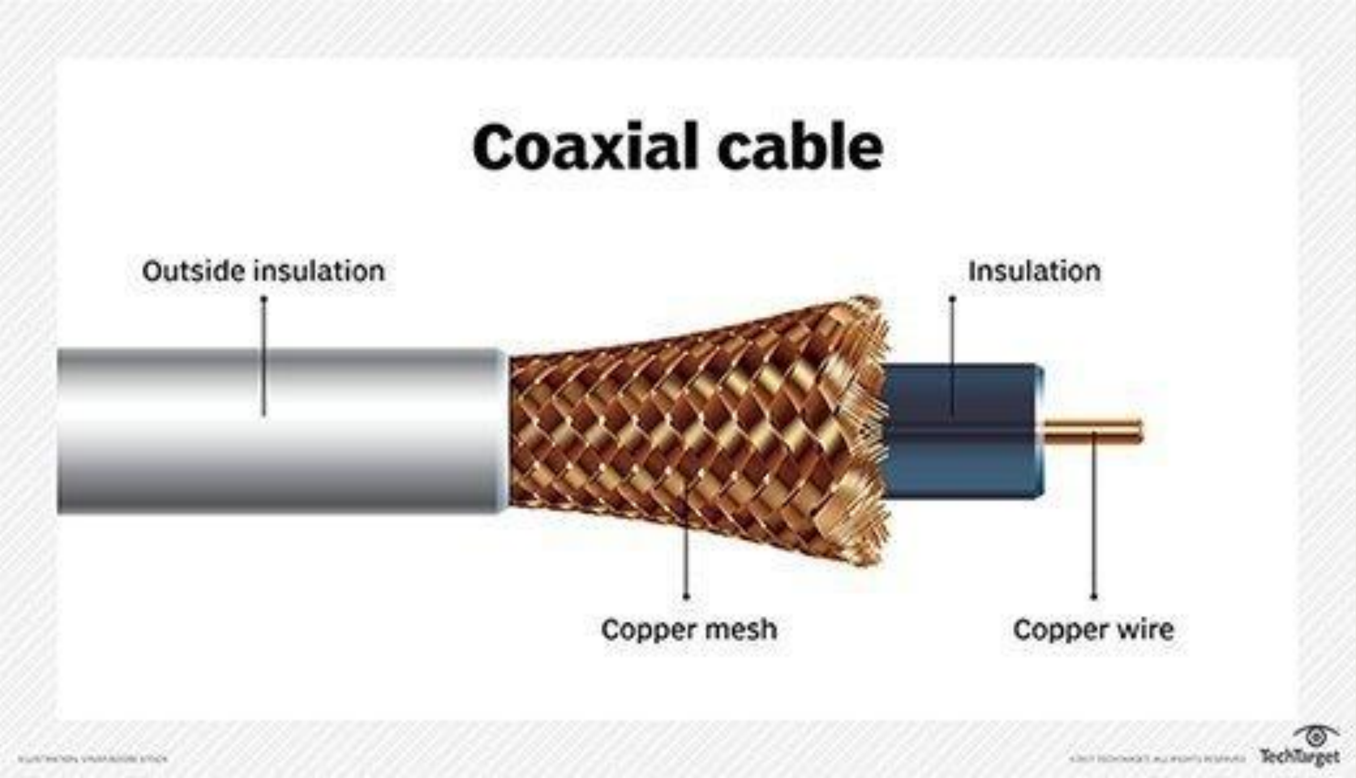
COAXIAL CABLE

- Another widely used transmission medium is the **coaxial cable**, commonly called **"coax."**
- Compared to unshielded twisted-pair cables, coaxial cables offer **better shielding and higher bandwidth**, allowing data to be transmitted over **longer distances at higher speeds**.

- There are two main types of coaxial cables in use:

- **50-ohm coaxial cable-** used for digital data transmission
- **75-ohm coaxial cable-** used for analog signals and cable television.

- A coaxial cable has a **layered structure**.
- At the center is a solid copper core that carries the signal.



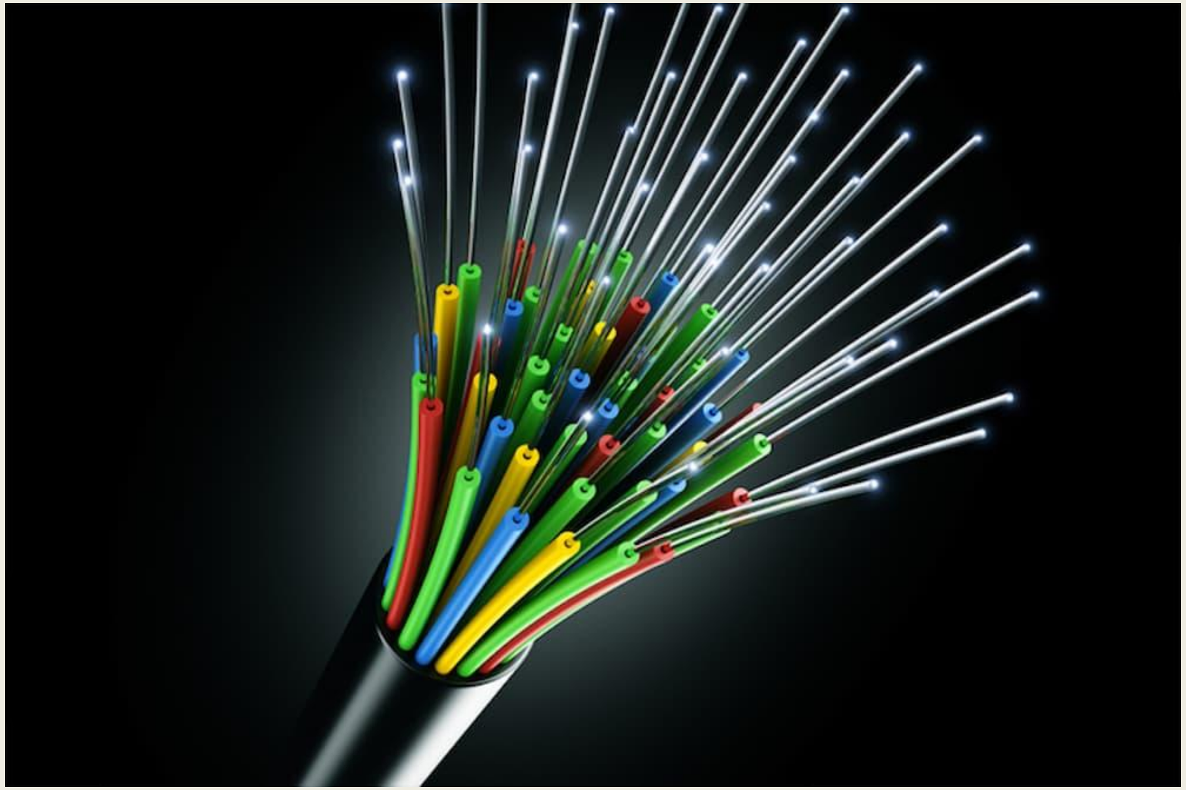
- This core is surrounded by an **insulating layer**, which is then enclosed by a **cylindrical outer conductor**, usually made of braided copper mesh.
- Finally, the entire cable is covered with a **protective plastic sheath**.

- This design provides strong shielding against external interference.

- Due to its construction, coaxial cable provides **excellent noise immunity** and supports **very high bandwidths**.
- Modern coaxial cables can support frequencies of up to **6 GHz**, allowing multiple signals to be transmitted simultaneously.
- Although coaxial cables were once widely used for long-distance telephone communication, they have largely been replaced by **fiber-optic cables** for long-haul links.
- However, coaxial cables are still extensively used in **cable television networks, metropolitan area networks (MANs), and broadband Internet connections to homes**.

FIBER-OPTIC

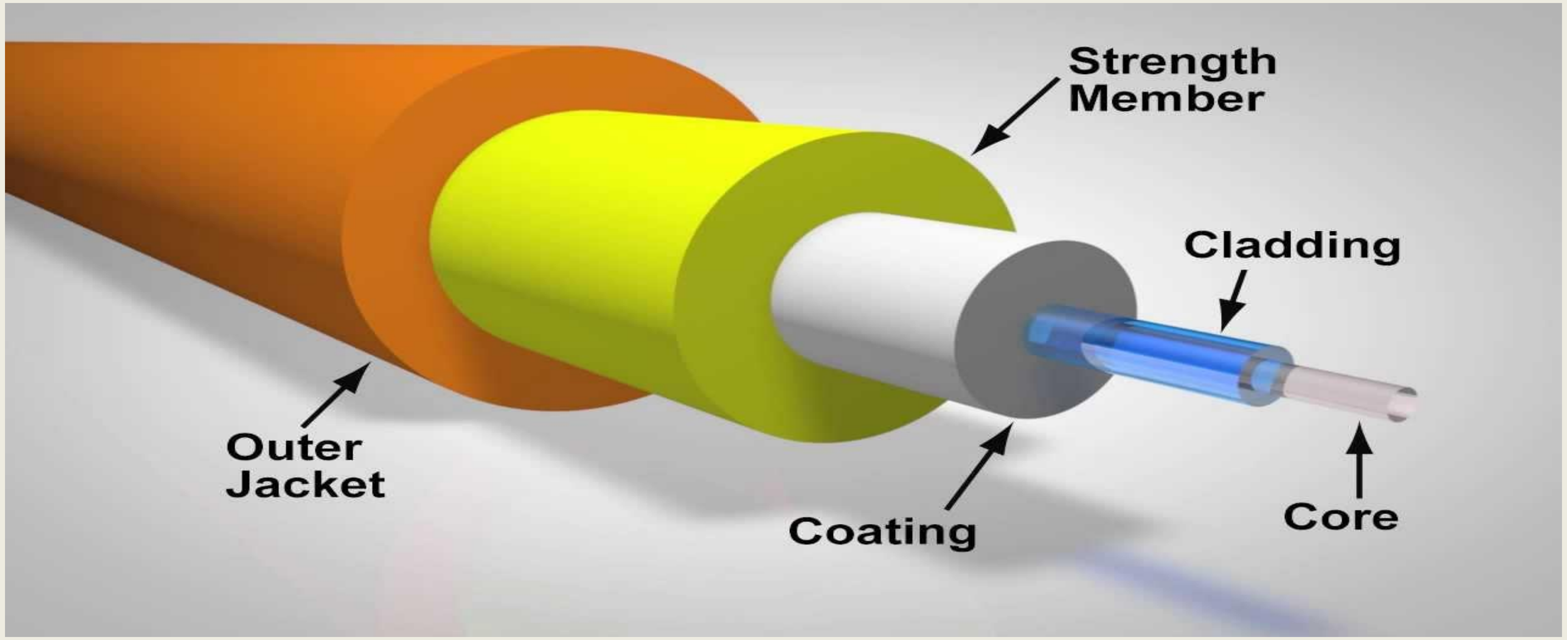
- Computer technology has improved rapidly over the years. Early computers ran at only a few megahertz, while modern systems operate at several gigahertz with multiple cores.



- Unlike CPUs, which are nearing physical limits, **fiber-optic communication still has huge unused potential**.
- Fiber cables can theoretically support extremely high data rates, reaching tens of terabits per second.
- To increase speed, multiple signals are often sent in parallel through the same fiber.

- Fiber optics are commonly used in **Internet backbones, high-speed local networks, and fiber-to-the-home connections**.

- A fiber-optic system has three main parts:
  - **Light source**
  - **Glass fiber**
  - **Detector**



- Data is sent as light pulses, where light represents a binary 1 and no light represents a binary 0.
- The receiver converts these light pulses back into electrical signals.

There are two main types of fibers.

- **Multimode fiber** allows light to travel along multiple paths and is suitable for short distances.
- **Single-mode fiber** allows light to travel in a single straight path and supports much higher speeds over longer distances.
- Single-mode fiber is more expensive but can transmit data at very high speeds over long distances without amplification.
- Overall, **fiber-optic technology provides extremely high speed and reliability**, making it the backbone of modern communication networks.
- Even though installation costs can be high, especially for connecting homes and small offices.

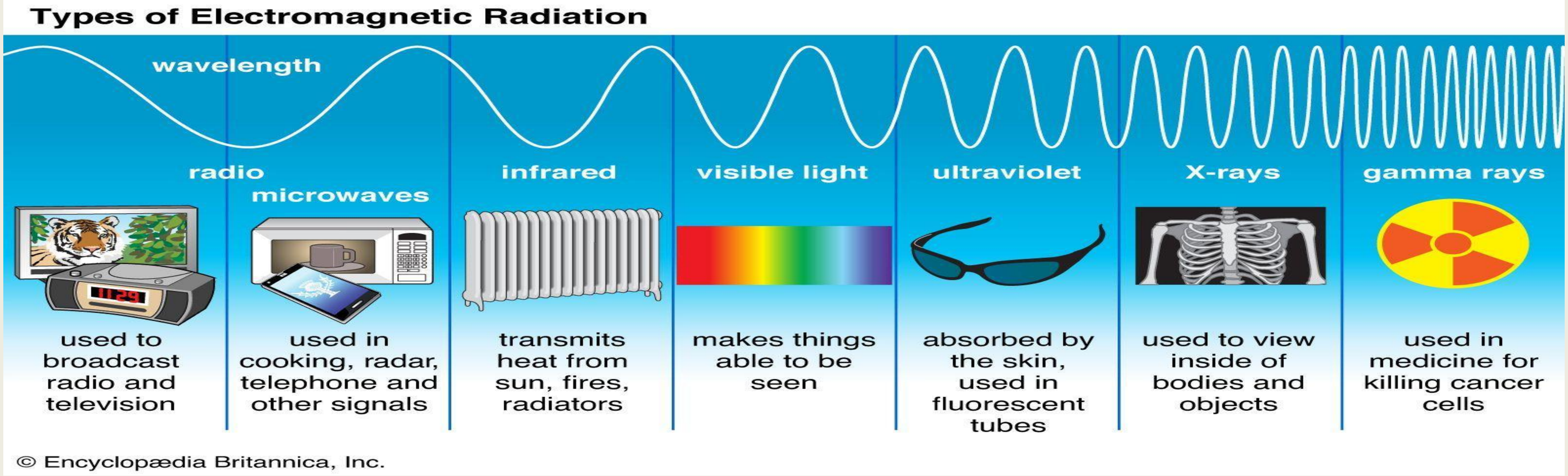
Wireless Transmission

- Today, many devices such as **laptops, smartphones, smartwatches, and even smart home appliances** use wireless connectivity to communicate with each other and with networks.
- Wireless communication allows these devices to send and receive data without physical cables.
- Wireless communication is useful not only for mobile users but also for **fixed devices** in certain situations.
- For example, when it is difficult or expensive to install wired connections such as in **mountainous areas, forests, remote locations** wireless networks can be a better solution.

Electromagnetic Spectrum

- When **electrons move**, they produce **electromagnetic waves** that can travel through space, even through a vacuum.
- These waves were first **predicted by James Clerk Maxwell in 1865** and later **experimentally observed by Heinrich Hertz in 1887**.
- All wireless communication works using these electromagnetic waves.
- The **frequency (f)** of a wave is the number of oscillations per second and is measured in **Hertz (Hz)**.
- The **wavelength (λ)** is the distance between two consecutive peaks of the wave.
- When an antenna of suitable size is connected to an electrical circuit, it can efficiently **transmit and receive electromagnetic waves**.
- Frequency, wavelength, and speed of light are related by the equation **$f = c / \lambda$**.

- The **electromagnetic spectrum** includes radio waves, microwaves, infrared, and visible light.
- These parts of the spectrum are commonly used for communication by changing the **amplitude, frequency, or phase** of the waves.
- Higher-frequency waves such as X-rays and gamma rays are not used because they are difficult to control, do not pass well through buildings, and are harmful to living beings.



Frequency Hopping Spread Spectrum

- **Frequency Hopping Spread Spectrum (FHSS)** is a wireless transmission technique in which the transmitter rapidly switches, or *hops*, between different frequencies many times per second.
- Because the signal does not stay on one frequency for long, it is **very difficult to detect or jam**, which is why this method was originally popular in **military communication**.
- This technique is also **robust and reliable**. It reduces problems caused by fading, where signals take multiple paths and interfere with each other.
- It also protects against **narrowband interference**, since even if one frequency is noisy or blocked, communication continues on other frequencies.
- Due to these advantages, frequency hopping works well in **crowded frequency bands**, such as the ISM bands, and is used in technologies like **Bluetooth** and older versions of **Wi-Fi (802.11)**.

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Direct Sequence Spread Spectrum

- **Direct Sequence Spread Spectrum (DSSS)** is a spread spectrum technique in which the original data signal is multiplied by a high-rate **code sequence**.
- This process spreads the signal over a **much wider frequency band** than the original data would normally require.
- A major advantage of DSSS is that **multiple users can share the same frequency band** at the same time by using **different code sequences**.
- This method is known as **Code Division Multiple Access (CDMA)**. Because each receiver knows its own code, it can extract the intended signal while treating others as noise.

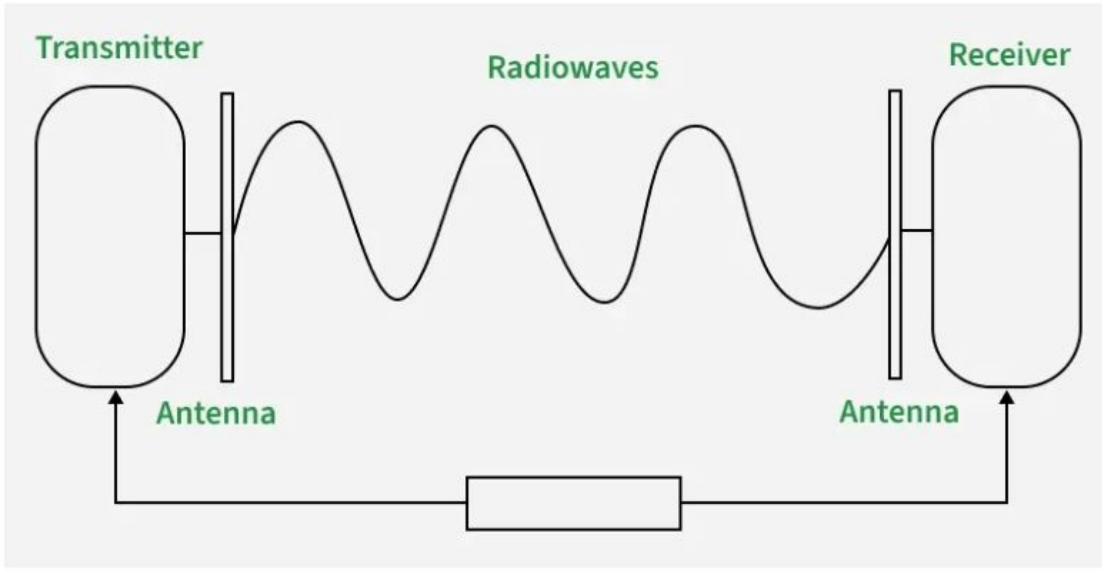
Ultra-Wideband (UWB) Communication

- Ultra-Wideband (UWB) communication transmits information using **very short, low-energy pulses** instead of continuous waves.
- These rapid pulses spread the signal over a **very wide range of frequencies**, resulting in extremely high bandwidth.
- A signal is considered **UWB** if it has a bandwidth of **at least 500 MHz** or **20% of its center frequency**.
- Because of this large bandwidth, UWB can support **very high data rates**, often reaching **hundreds of megabits per second**.
- UWB is **highly resistant to interference** from other narrowband signals since the information is spread across many frequencies.
- At the same time, the power at any single frequency is very low, so UWB **does not significantly interfere with other wireless systems** operating in the same spectrum.

USING THE SPECTRUM FOR TRANSMISSION

Radio Transmission

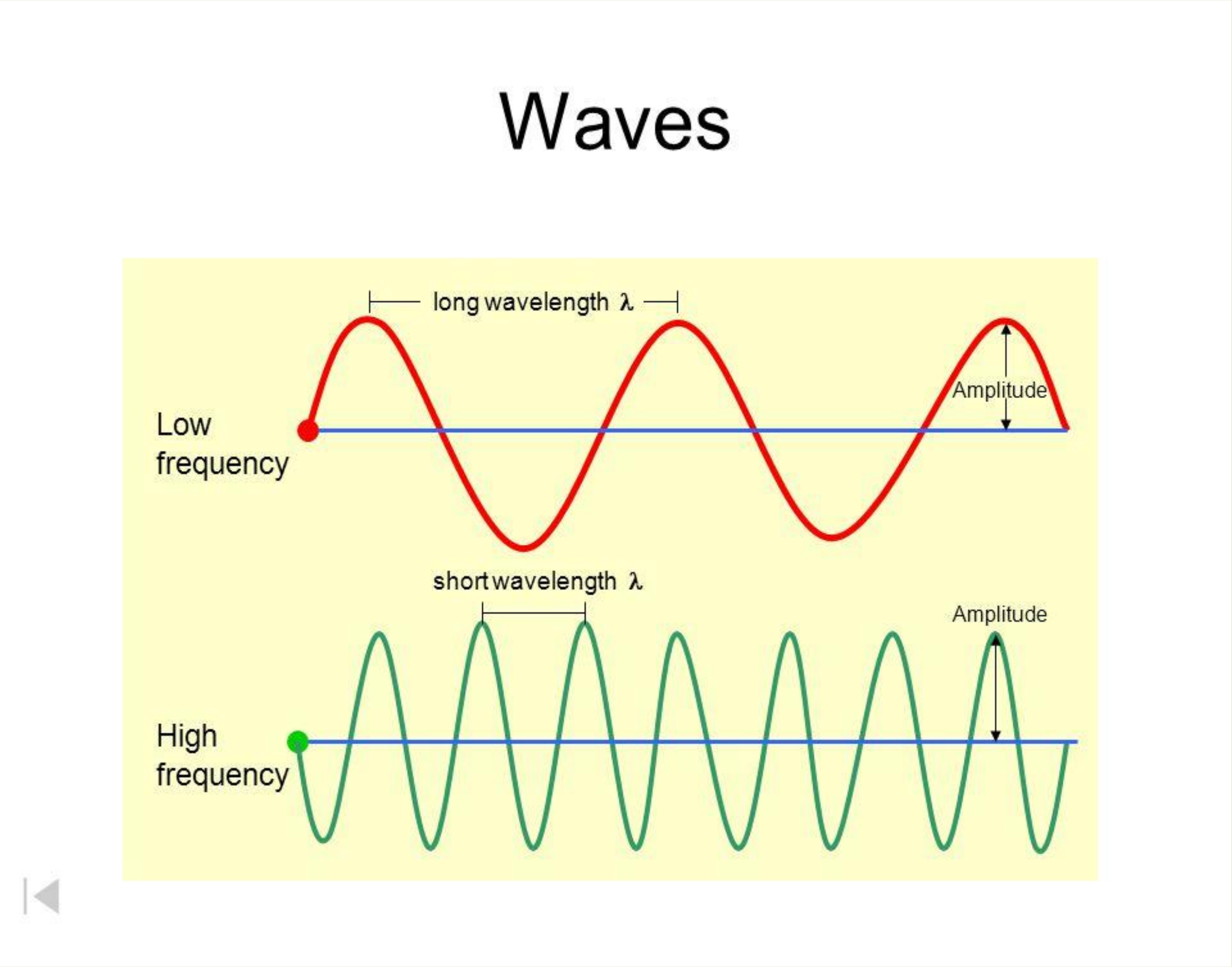
- Radio frequency (RF) waves are widely used for communication because they are **easy to generate**, can **travel long distances**, and can **penetrate buildings**.



Radiowaves

- They are also **omnidirectional**, meaning they spread out in all directions from the transmitter, so the sender and receiver do not need precise alignment.
- Since radio waves spread everywhere, **unwanted receivers can also pick up the signal**, and interference can occur.

- The behavior of radio waves depends strongly on their **frequency**.



- **Low-frequency radio waves** pass through obstacles easily but lose power rapidly as distance increases. This loss of signal strength with distance is called **path loss**.
- **High-frequency radio waves** usually travel in straight lines, reflect off objects, and are more affected by rain and obstacles. They can suffer from reflections that cause signal variation.

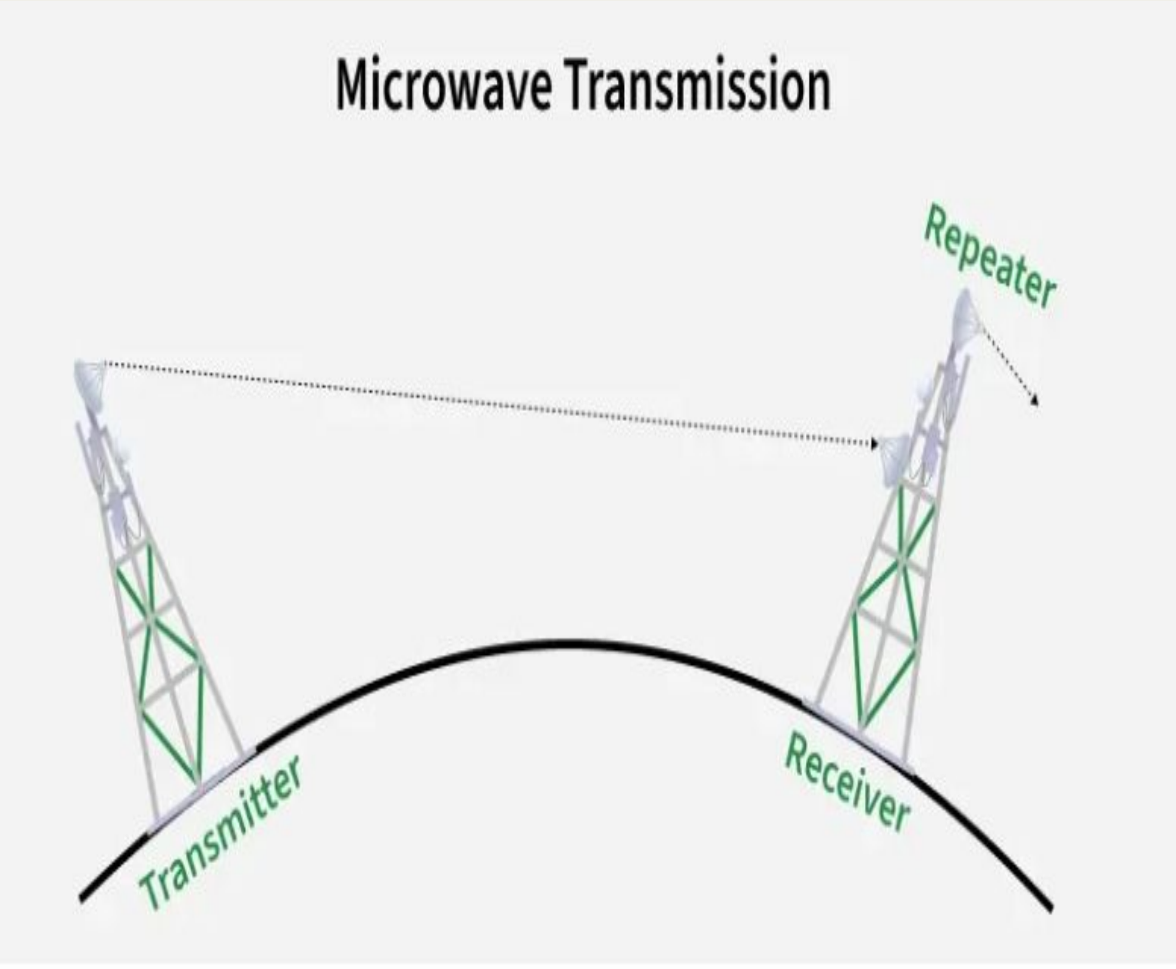
- Compared to guided media like twisted pair or fiber, radio waves behave differently.
- In cables, signal strength decreases steadily with distance. In radio transmission, signal strength drops significantly every time the distance doubles.
- Because radio waves can travel far and interfere with many users.
- In **very low (VLF), low (LF), and medium frequency (MF) bands**, radio waves follow the surface of the Earth and can travel hundreds of kilometers.
- In **high frequency (HF) and very high frequency (VHF) bands**, radio waves can reflect off the **ionosphere** and return to Earth, allowing long-distance communication.

Microwave Transmission

- It is used at **frequencies above 100 MHz**; waves travel in **straight lines**
- By concentrating energy in one direction, a **higher signal-to-noise ratio** is achieved, but this requires the **transmitting and receiving antennas to be precisely aligned**.
- As the microwaves are directional, **multiple transmitters and receivers can operate close to each other without interference**, as long as proper spacing is maintained.
- These are widely used earlier for **long-distance telephone networks** before fiber optics.
- Requires **repeaters (towers)** at intervals due to Earth's curvature; taller towers allow greater spacing (≈80 km for 100 m towers).

Advantages:

- Microwave communication is widely used for **long-distance telephony, mobile networks, and television distribution**.
- Its major advantages are that **no physical cables are needed**, it is **relatively inexpensive**, and it avoids the need for right-of-way permissions required for laying fiber.
- This made microwave systems an attractive alternative to wired networks, especially in difficult terrain or congested urban areas.



Microwave Transmission

Infrared transmission

- Infrared transmission is mainly used for short-range communication, such as in television and multimedia remote controls.
- It is inexpensive, easy to build, and relatively directional, but it has an important limitation
- Infrared waves cannot pass through solid objects, so communication works only when there is a clear line of sight.
- This limitation also acts as an advantage because infrared systems in one room do not interfere with systems in nearby rooms, and the risk of eavesdropping is low.

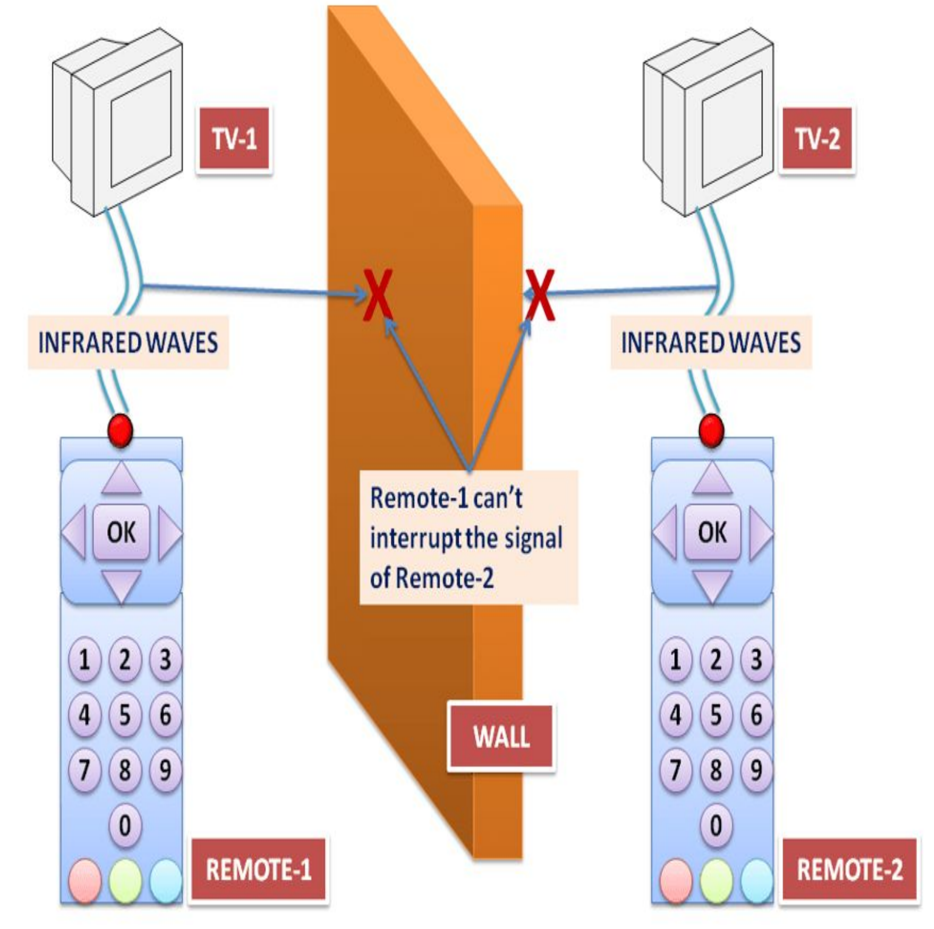


Figure 3: Infrared Waves Transmission

Light transmission

- Light transmission, also called **unguided optical or free-space optical communication**, has been used for centuries, Today, it is commonly used to connect LANs between nearby buildings using rooftop lasers.
- This method provides **very high bandwidth at low cost** and is relatively **secure**, since tapping a narrow laser beam is difficult.
- It is easy to install and does not require government licensing, unlike microwave transmission.
- Laser communication is **highly directional**, so precise alignment is needed, and environmental factors such as **wind, temperature changes, rain, fog, and atmospheric turbulence** can disrupt the signal.
- Light-based communication is gaining interest through **visible light communication (VLC)**, where data is transmitted using LEDs by rapidly switching light patterns faster than human perception.